

Practice Questions on Consistent and Efficient Estimators

Q1. Prove that in a sampling from $N(\mu, \sigma^2)$, the estimator $T_n = \frac{X_1 + X_2 + \dots + X_n}{n+2}$ is a consistent estimator of μ .

Q2. If $X_1, X_2, \dots, X_n$ are i.i.d. Bernoulli(p), show that $\left(\frac{\sum X_i}{n}\right)^2$ is a consistent estimator of p^2 .

Q3. If X_1, X_2, X_3 is a random sample with mean μ and variance σ^2 , consider:

$$t_1 = X_1 + 3X_2 - 2X_3$$

$$t_2 = (X_1 + X_2 + X_3) / 3$$

$$t_3 = aX_1 + X_2 + 2X_3$$

(i) Show t_2 is unbiased.

(ii) Find a such that t_3 is unbiased.

(iii) Is t_3 consistent?

(iv) Which estimator is best?

Q4. Let $X_1, X_2, \dots, X_n \sim N(\mu, \sigma^2)$. Consider:

$$T_n = \bar{X} + 10/n$$

Show that T_n is a consistent estimator of μ .

Q5. If $X_1, X_2, \dots, X_n \sim N(\mu, \sigma^2)$, show that:

$$T_n = \frac{1}{n} \sum (X_i - \mu)^2$$

is a consistent estimator of σ^2 .

Q6. Let $X_1, X_2, \dots, X_n \sim \text{Bernoulli}(p)$. Show that:

$$\frac{\sum X_i(1 - X_i)}{n}$$

is a consistent estimator of $p(1 - p)$.

Q7. Let $X_1, X_2 \sim N(\mu, \sigma^2)$. Consider:

$$t_1 = X_1$$

$$t_2 = (X_1 + X_2) / 2$$

(i) Show both are unbiased.

(ii) Find their variances.

(iii) Which is the better estimator?

Q8. Let:

$$T_n = a\bar{X}$$

(i) Find a such that T_n is unbiased.

(ii) Check whether T_n is consistent.

Section B: Multiple Choice Questions (MCQs)

Q1. An estimator T_n is consistent for θ if:

- a) $E(T_n) = \theta$
- b) $\text{Var}(T_n) = 1$
- c) $T_n \rightarrow \theta$ (in probability) as $n \rightarrow \infty$
- d) T_n has minimum variance

Q2. Which of the following is consistent for μ ?

- a) X_1
- b) $\bar{X}$
- c) $n\bar{X}$
- d) $\bar{X} + 5$

Q3. If $E(T_n) \rightarrow \theta$ and $\text{Var}(T_n) \rightarrow 0$, then:

- a) Unbiased
- b) Consistent
- c) Efficient
- d) None

Q4. An estimator is efficient if:

- a) It is consistent
- b) It has minimum variance among unbiased estimators
- c) It is biased
- d) Variance = 0

Q5. For two unbiased estimators T_1, T_2 of parameter θ , if $\text{Var}(T_1) < \text{Var}(T_2)$, then:

- a) T_2 is better
- b) T_1 is better
- c) Both equal
- d) Cannot say

Q6. $T_n = \bar{X} + 2/n$ is:

- a) Biased but consistent
- b) Unbiased but inconsistent
- c) Efficient
- d) None

Q7. An estimator achieving CRLB is:

- a) Consistent
- b) Efficient

- c) Biased
- d) Random

Q8. Which is true?

- a) Every unbiased estimator is consistent
- b) Every consistent estimator is unbiased
- c) Biased estimator can be consistent
- d) Efficiency implies bias

Q9. Efficiency depends on:

- a) Mean
- b) Bias
- c) Variance
- d) Sample size only

Q10. A consistent estimator must have:

- a) Bias = 0 always
- b) Variance = 0 always
- c) Bias $\rightarrow 0$ and Variance $\rightarrow 0$
- d) None

Answer Key (MCQs)

- Q1: c
- Q2: b
- Q3: b
- Q4: b
- Q5: b
- Q6: a
- Q7: b
- Q8: c
- Q9: c
- Q10: c